PADHAKU
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CHAPTER 7

The Mathematics of Maybe: Introduction to Probability

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🤔 Think and Reflect · 7.1.1 What is Randomness?

? **In a cricket match, a coin is tossed to decide which team will bat first. This is considered a fair method. Can you explain why?**

✔ ANSWER

- A fair coin has only two outcomes, heads or tails, and both are **equally likely** (probability 1/2 each).
- No captain can know or control the result in advance.

So each team has the **same chance (1/2) of winning the toss**, which makes it fair.

? **Ask your friend to predict the outcome of a ₹1 coin you toss. Do you see that your friend could guess heads or tails but could not know for certain?**

💡 SAMPLE ANSWER

Yes. The possible results are known: {Heads, Tails}.

But one toss cannot be predicted; the guess is right only about half the time.

All results are known, but each single toss is unpredictable — this is randomness.

Exercise Set 7.1

1 (i) **Rank on a scale from 0 (Impossible) to 1 (Certain) and give reasons:**
The next Monday will come after Sunday.

✔ ANSWER

Monday always comes right after Sunday.

Certain (probability 1).

1 (ii) **It will snow in Mumbai in July.**

💡 SAMPLE ANSWER

Mumbai is a hot coastal city and July is the rainy (monsoon) season; it has never snowed there.

Impossible (probability 0, or almost 0).

1 (iii) **An elephant will walk through your classroom today.**

💡 SAMPLE ANSWER

Elephants are not found in schools; this almost never happens.

Impossible / extremely unlikely (close to 0).

1 (iv) **You will greet at least one friend at school tomorrow.**

💡 SAMPLE ANSWER

On a normal school day we usually meet and greet friends (unless it is a holiday or you are absent).

More likely (close to 1).

📝 Exercise Set 7.2

1 (i) **A sample of 30 sweets: 10 red | 8 green | 7 yellow | 5 blue.**

Calculate the probability that a randomly picked sweet from the sample is green.

✔ ANSWER

Green sweets = 8, total = 30

P(green) = 8/30 = **4/15 ≈ 0.267**

1 (ii) **If there are 600 sweets in total in the large bag, estimate how many are likely to be yellow.**

✔ ANSWER

P(yellow) from the sample = 7/30

Estimated yellow sweets = 7/30 × 600 = 7 × 20 = **140**

2 (i) **Sample of 40 students: Science 14 | Arts 11 | Sports 9 | Debate 6. School has 800 students.**

What is the probability that a randomly chosen student from the sample prefers the

Arts Club?

✔ ANSWER

P(Arts) = 11/40 = **0.275**

2 (ii) **Estimate how many students in the whole school are likely to prefer the Sports Club.**

✔ ANSWER

P(Sports) = 9/40

Estimate = 9/40 × 800 = 9 × 20 = **180 students**

3 **Toss a coin 20 times and record each result. (i) How many heads? (ii) How many tails? (iii) Experimental probability of heads? (iv) If you toss once more, what is the probability of tails?**

💡 SAMPLE ANSWER

Results depend on your own tosses. Example: 11 heads, 9 tails.

(i) Heads = **11** (example)

(ii) Tails = 20 − 11 = **9** (example)

(iii) Experimental P(heads) = 11/20 = **0.55** (example)

(iv) The next toss does not depend on earlier tosses: P(tails) = **1/2**

4 **Toss a paper cup into the air 100 times. Record whether it lands on its bottom, upside**

down on its top, or on its side. Assign probabilities by experimental probability.

💡 SAMPLE ANSWER

Example record:

Landing	Times	Experimental probability
Bottom	18	18/100 = 0.18
Top (upside down)	22	22/100 = 0.22
Side	60	60/100 = 0.60

Rule: P = number of times it landed that way ÷ 100. The three probabilities add up to **1**.

Usually the cup lands on its side most often, so these outcomes are **not equally likely**.

5 **What is the probability of getting an even number when rolling a fair 6-sided die?**

✔ ANSWER

S = {1, 2, 3, 4, 5, 6}; even numbers = {2, 4, 6} → 3

P(even) = 3/6 = **1/2**

6 (i) **You roll a die 12 times and get a '3' three times. What is the experimental probability of rolling a '3'?**

✔ ANSWER

Experimental P(3) = 3/12 = **1/4 = 0.25**

6 (ii) **What is the theoretical probability of rolling a '3'?**

✔ ANSWER

One favourable outcome out of 6

P(3) = **1/6 ≈ 0.167**

6 (iii) **Why might these probabilities be different? What would you expect if you roll the die 60, 600 or 6000 times?**

✔ ANSWER

12 rolls is a small number of trials, so chance can give a result far from the ideal value.

As the number of rolls increases (60, 600, 6000), the experimental probability gets **closer and closer to 1/6** (about 10, 100, 1000 threes).

This is the **Law of Large Numbers**.

📝 Exercise Set 7.3

1 **When a single 6-sided die is rolled, what is the total number of possible outcomes in the sample space?**

✔ ANSWER

S = {1, 2, 3, 4, 5, 6}

6 outcomes

2 (i) **Write the sample space: Rolling a die and tossing a coin together.**

✔ ANSWER

S = {1H, 2H, 3H, 4H, 5H, 6H, 1T, 2T, 3T, 4T, 5T, 6T}

n(S) = 12

2 (ii) **Choosing a random integer between −5 and +5.**

💡 SAMPLE ANSWER

S = {−4, −3, −2, −1, 0, 1, 2, 3, 4}, **n(S) = 9**

(If −5 and +5 are also included: S = {−5, −4, ..., 4, 5}, n(S) = 11.)

2 (iii) **A box contains 5 green and 7 red balls. One ball is drawn at random.**

✔ ANSWER

By colour: **S = {Green, Red}**

If each ball is counted separately: S = {G1, G2, G3, G4, G5, R1, R2, ..., R7}, n(S) = 12

3 (i) **Snacks: Samosa, Pakora, Bhaji; drinks: Chai or Lassi. List the sample space of all snack and drink combinations.**

✔ ANSWER

S = {(Samosa, Chai), (Samosa, Lassi), (Pakora, Chai), (Pakora, Lassi), (Bhaji, Chai), (Bhaji, Lassi)}

n(S) = 3 × 2 = 6

3 (ii) **List the event 'Selecting Samosa as a snack.'**

✔ ANSWER

E = {(Samosa, Chai), (Samosa, Lassi)}

🤔 Think and Reflect · 7.4 Tree diagrams

? **(Tossing a fair coin two times.) Can you calculate the probability of getting one head and one tail?**

✔ ANSWER

S = {HH, HT, TH, TT}, n(S) = 4

Favourable outcomes: HT, TH → 2

P(one head and one tail) = 2/4 = **1/2 = 0.5 or 50%**

Exercise Set 7.4

1 (i) **Basket A: 1 apple, 2 oranges. Basket B: 1 banana, 1 mango. One fruit is picked from each basket. Draw a tree diagram showing all possible pairs.**

✔ ANSWER

Call the oranges O1 and O2.

Start → Apple → Banana / Mango

Start → Orange 1 → Banana / Mango

Start → Orange 2 → Banana / Mango

First branches (Basket A): 3, each 1/3. Second branches (Basket B): 2, each 1/2.

6 paths = 6 possible pairs.

1 (ii) **List the sample space.**

✔ ANSWER

S = {(Apple, Banana), (Apple, Mango), (O1, Banana), (O1, Mango), (O2, Banana), (O2, Mango)}, n(S) = 6

(By fruit type: Apple–Banana, Apple–Mango, Orange–Banana, Orange–Mango.)

1 (iii) **What is the probability of picking one apple and one banana?**

✔ ANSWER

Favourable: (Apple, Banana) → 1 out of 6 equally likely outcomes

P = 1/3 × 1/2 = **1/6**

2 (i) **A box has 3 red, 4 black and 2 green pens. You pick a pen and put it back; then your friend does the same. What are the possible outcomes of the pen colours? Draw a tree diagram.**

✔ ANSWER

Each pick: P(Red) = 3/9 = 1/3, P(Black) = 4/9, P(Green) = 2/9.

Tree: Start → R, B, G (you); from each, again → R, B, G (friend).

Outcomes (you, friend): **RR, RB, RG, BR, BB, BG, GR, GB, GG** (9 outcomes).

2 (ii) **Use the tree diagram to find the probability that both of you pick pens of the same colour.**

✔ ANSWER

Multiply along each branch:

P(RR) = 3/9 × 3/9 = 9/81

P(BB) = 4/9 × 4/9 = 16/81

P(GG) = 2/9 × 2/9 = 4/81

P(same colour) = (9 + 16 + 4)/81 = **29/81 ≈ 0.36**

End-of-Chapter Exercises

1 (i) The probability of an impossible event is ____.

✔ ANSWER

0

1 (ii) The set of all possible outcomes of a random experiment is called the ____.

✔ ANSWER

sample space

1 (iii) The probability of an event that is certain to happen is ____.

✔ ANSWER

1

1 (iv) Tossing a fair coin has a probability of ____ for getting heads.

✔ ANSWER

1/2 (0.5)

2 In a survey of 50 students, 15 like football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____.

✔ ANSWER

relative frequency is 15/50 = **3/10 = 0.3**

3 Which experiments have equally likely outcomes? Explain. (i) A car starts or does not start. (ii) Tossing a fair coin once. (iii) Rolling a fair 6-sided die. (iv) Choosing a marble from a bag of 3 red and 7 blue marbles. (v) A baby is born: a boy or a girl.

✔ ANSWER

(i) **Not equally likely** – a car usually starts; it depends on its condition.

(ii) **Equally likely** – heads and tails each have probability 1/2.

(iii) **Equally likely** – each face has probability 1/6.

(iv) **Not equally likely** – P(red) = 3/10, P(blue) = 7/10.

(v) **Equally likely** (taken so) – a boy or a girl is each about 1/2.

4 (i) Two coins are tossed. What is the probability of getting at least one head?

✔ ANSWER

S = {HH, HT, TH, TT}

At least one head: HH, HT, TH → 3

P = **3/4**

4 (ii) **Cards numbered 1 to 10; one is drawn. Probability of an even number?**

✔ ANSWER

S = {1, 2, 3, ..., 10}, n(S) = 10

Even: {2, 4, 6, 8, 10} → 5

P = 5/10 = **1/2**

4 (iii) **A die is rolled once. Probability of a number greater than 4?**

✔ ANSWER

S = {1, 2, 3, 4, 5, 6}

Greater than 4: {5, 6} → 2

P = 2/6 = **1/3**

4 (iv) **A bag has 3 red, 2 blue and 1 green ball. One ball is picked. Probability that it is not red?**

✔ ANSWER

S = {R1, R2, R3, B1, B2, G}, n(S) = 6

Not red: B1, B2, G → 3

P = 3/6 = **1/2**

4 (v) **Three coins are tossed. Probability of exactly two heads?**

✔ ANSWER

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}, n(S) = 8

Exactly two heads: HHT, HTH, THH → 3

P = **3/8**

5 **A bag has 3 candies: strawberry, lemon and mint. Probability of picking strawberry?**

✔ ANSWER

S = {Strawberry, Lemon, Mint}

P(strawberry) = **1/3**

6 **2 shirts (red, blue) and 3 pants (jeans, khakis, shorts). List all outfits of one shirt and one pair of pants in a table.**

✔ ANSWER

Shirt	Pants	Outfit
Red	Jeans	Red shirt + Jeans
Red	Khakis	Red shirt + Khakis
Red	Shorts	Red shirt + Shorts
Blue	Jeans	Blue shirt + Jeans
Blue	Khakis	Blue shirt + Khakis
Blue	Shorts	Blue shirt + Shorts

2 × 3 = 6 outfits

7 (i) **A tyre company records distances before replacement in 1000 cases:**

Distance (km)	Less than 4000	4001 to 9000	9001 to 14000	More than 14000
Number of cases	20	210	325	445

Probability that a randomly chosen tyre lasts less than 4000 km?

✔ ANSWER

P = 20/1000 = **0.02**

7 (ii) **Between 4000 and 14000 km?**

✔ ANSWER

Cases = 210 + 325 = 535

P = 535/1000 = **0.535**

7 (iii) **More than 14000 km?**

✔ ANSWER

P = 445/1000 = **0.445**

8 (i) **The letters of 'PEACE' are on cards: P, E, A, C, E. Leela draws one. Probability that it is a P, E or C?**

✔ ANSWER

Total cards = 5

P, E, E, C → 4 cards

P = **4/5**

8 (ii) **Probability that it is not an E?**

✔ ANSWER

Not E: P, A, C → 3

P = **3/5**

***9** **An arrow stops at one of 1, 2, 3, 4, 5, 6, 7, 8 (equally likely). Probability that it points at (i) 8 (ii) an odd number (iii) a number greater than 2 (iv) a number less than 9 (v) a multiple of 3?**

✔ ANSWER

n(S) = 8

(i) {8} → **1/8**

(ii) {1, 3, 5, 7} → 4/8 = **1/2**

(iii) {3, 4, 5, 6, 7, 8} → 6/8 = **3/4**

(iv) all 8 numbers → 8/8 = **1** (certain)

(v) {3, 6} → 2/8 = **1/4**

***10** **A basket has 4 red and 5 blue balls. One ball is drawn and laid aside, then a second is drawn. Draw a tree diagram. (i) Probability of red then blue? (ii) Probability of 2 blue balls?**

✔ ANSWER

Tree (9 balls at first, 8 left for the second draw):

- First Red (4/9) → second Red (3/8) or Blue (5/8)
- First Blue (5/9) → second Red (4/8) or Blue (4/8)

Outcome	Probability
RR	4/9 × 3/8 = 12/72
RB	4/9 × 5/8 = 20/72
BR	5/9 × 4/8 = 20/72
BB	5/9 × 4/8 = 20/72

(Check: 12 + 20 + 20 + 20 = 72 → total 1.)

(i) P(red then blue) = 20/72 = **5/18**

(ii) P(2 blue) = 20/72 = **5/18**

***11** **I throw a pair of 6-sided dice. Write down an event that has a probability of 0 and one that has a probability of 1.**

💡 SAMPLE ANSWER

Sums can only be 2 to 12.

Probability 0: ‘The sum is 13’ (or ‘the sum is 1’) — **impossible**.

Probability 1: ‘The sum is between 2 and 12’ (or ‘each die shows a number less than 7’) — **certain**.

***12 (i)** **Two dice are rolled. Probability that the sum is a prime number greater than 5?**

✔ ANSWER

S = {(1,1), (1,2), ..., (6,6)}, n(S) = 6 × 6 = 36

Primes greater than 5 (sum ≤ 12): 7 and 11

Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) → 6

Sum 11: (5,6), (6,5) → 2

P = 8/36 = **2/9**

***12 (ii)** **A bag has 4 red, 3 green and 2 blue balls. Two balls are drawn without replacement. Probability that both are of different colours?**

✔ ANSWER

Total pairs of balls = 9 × 8 ÷ 2 = 36

Same colour pairs: red 4×3÷2 = 6, green 3×2÷2 = 3, blue 2×1÷2 = 1 → 10

Different colours = 36 − 10 = 26

P = 26/36 = **13/18**

***12 (iii)** **Three coins are tossed. Probability that the first coin shows heads and exactly two heads occur in total?**

✔ ANSWER

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}, n(S) = 8

First is H and exactly two H: HHT, HTH → 2

P = 2/8 = **1/4**

***12 (iv)** **A four-digit number is formed using 1, 2, 3 and 4 with no repetition. Probability that the number is even?**

✔ ANSWER

Total numbers = 4 × 3 × 2 × 1 = 24

Even: last digit 2 or 4 → 2 × (3 × 2 × 1) = 12

P = 12/24 = **1/2**

***12 (v)** **A test has 3 questions, each with 4 options (A, B, C, D), one correct. A student guesses. Probability of exactly 2 correct?**

✔ ANSWER

Total ways to answer = 4 × 4 × 4 = 64

Choose which 2 are correct: 3 ways (Q1Q2, Q1Q3, Q2Q3)

Each correct one: 1 way; the wrong one: 3 ways

Favourable = 3 × 1 × 1 × 3 = 9

P = **9/64**

***13 (i)** **A box has 4 balls numbered 1 to 4. A ball is drawn and recorded, returned, and a second ball is drawn and recorded. Record the sample space using a tree diagram.**

✔ **ANSWER**

Tree: first draw 1, 2, 3, 4; from each, second draw 1, 2, 3, 4.

S = {(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)}

***13 (ii)** **A ball is drawn and recorded; without replacing it, a second ball is drawn and recorded.**

✔ **ANSWER**

Tree: first draw 1, 2, 3, 4; from each, only the 3 other numbers.

S = {(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)}

***13 (iii)** **What are the sizes of these two sample spaces?**

✔ **ANSWER**

(i) 4 × 4 = **16**

(ii) 4 × 3 = **12**

***14** **List the sample space for tossing a coin and drawing a card from 6 cards numbered 1 to 6.**

✔ ANSWER

S = {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}

n(S) = 2 × 6 = 12

***15** **Three coins are tossed and the number of heads is recorded. Which list is a sample space? (i) {1, 2, 3} (ii) {0, 1, 2} (iii) {0, 1, 2, 3, 4} (iv) {0, 1, 2, 3}**

✔ ANSWER

With 3 coins, the number of heads can be 0, 1, 2 or 3.

(iv) {0, 1, 2, 3} is the sample space.

(i) misses 0 (all tails).

(ii) misses 3 (all heads).

(iii) has 4, which is impossible with 3 coins.

A sample space must contain every possible outcome and nothing else.

***16** **A drop is dropped at random on a 3 m × 2 m rectangle (Fig. 7.8). What is the probability that it lands inside the circle of diameter 1 m?**

✔ ANSWER

Area of rectangle = 3 × 2 = 6 m^2

Radius = 1/2 m; area of circle = $\pi \times (1/2)^2 = \pi/4$ m^2

P = $(\pi/4) \div 6 =$ **$\pi/24 \approx 0.131$** (using $\pi \approx 3.14$)

🎉 **Chapter 7 done! Revise the answers once more tomorrow.**